



REVIEW CHALLENGE

Problem statement

AI-LECTURE-TO-LEARNING-MATERIAL-GENERATOR

Advanced

General

AI Lecture-to-Learning Material Generator

Agentic AI

INDUSTRY

Education / EdTech / Higher Education

PLATFORM / TECH

Agentic AI, Generative AI, LLMs, NLP, Speech-to-Text, Document Processing, Web Application, Recommendation Systems, Data Analytics

SCOPE

Lecture Content Transformation / Learning Material Generation / Personalized Learning / Educational Content Management

DESCRIPTION

Develop an AI-powered platform that converts lecture content into structured learning materials. The system should accept lecture recordings, transcripts, notes, presentations, or other educational content and automatically extract the important concepts and organize them into useful study resources.

The platform should generate materials such as lecture summaries, structured notes, key concepts, question-and-answer sets, quizzes, flashcards, and revision resources while preserving the context and important information from the original lecture. It should allow students and educators to review, edit, and organize the generated learning materials.

KEY DELIVERABLES

- ☒ Lecture content upload and management
- ☒ Support for audio/video recordings
- ☒ Transcript and text input
- ☒ Speech-to-text processing
- ☒ Lecture transcription



Overview



My Team

- ✓ Key-point identification
- ✓ AI-generated lecture summaries
- ✓ Structured study notes
- ✓ Chapter/topic-wise organization
- ✓ Important terminology extraction
- ✓ Key concept explanations
- ✓ Automatic question generation
- ✓ Question-and-answer generation
- ✓ Quiz generation
- ✓ Multiple-choice questions
- ✓ Flashcard generation
- ✓ Revision material generation
- ✓ Learning-objective extraction
- ✓ Difficulty-level configuration
- ✓ Personalized learning-material generation
- ✓ Natural-language interaction with lecture content
- ✓ AI-powered lecture Q&A
- ✓ Search across lecture materials
- ✓ Source/context references to the original lecture
- ✓ Student editing and annotation
- ✓ Educator review and editing
- ✓ Saved learning-material library
- ✓ Lecture and subject organization
- ✓ Dashboard for students and educators
- ✓ Progress and revision tracking
- ✓ Export/download of generated materials
- ✓ Notifications and revision reminders
- ✓ Admin/content management
- ✓ Analytics on generated learning materials
- ✓ Human review and correction controls



Overview



My Team

 **Not confirmed yet**

You can confirm only one statement, and it can't be changed afterwards.

Confirm this statement

 **Copy reference**

COMPETITION DETAILS

 **OCT 3RD & 4TH 2026**

Hackathon dates

 **24HRS HACKATHON**

Offline event

 **2–6 Members**

Eligible team size

 **MRDU Campus**

Maisammaguda, Dulapally